Chapter 7

Average

An average or an arithmetic mean of given data is the sum of the given observations divided by number of observations.
For example: If we have to find out the average of 10, 15, 25 and 30, then the required average will be equal to

$$\frac{10+15+25+30}{4} = \frac{80}{4} = 20$$

Similarly, average of 30,10 and 50

$$= \frac{30+10+50}{3} = \frac{90}{3} = 30$$

Therefore, we can write the formula

$$Average(A) = \frac{\text{Sum of given observations}(S)}{\text{Number of Observation}(N)}$$

$$\therefore \quad A = \frac{S}{N}$$

7.1 Properties of Average

1. Average of a given data is less than the greatest observation and greater than the smallest observation of the given data.

   For examples: Average of 3,7,9 and 13

   $$= \frac{3+7+9+13}{4} = \frac{32}{4} = 8$$

   Clearly, 8 is less than 13 and greater than 3.

2. If the observations of given data are equal, then the average will also be the same as observations.

For example: Average of 6, 6, 6 and 6 will be 6 because

$$\frac{6+6+6+6}{4} = \frac{24}{4} = 6$$

3. If 0 (zero) is one of the observation of a given data, then that 0 (zero) will also be included while calculating average.
   For example: Average of 3, 6 and 0 is 3 because

$$\frac{0+3+6}{3} = \frac{9}{3} = 3$$

Note

★ If all the numbers get increased by a, then their average must be increased by a

★ If all the numbers get decreased by a, then their average must be decreased by a

★ If all the numbers are multiplied by a, then their average must be multiplied by a

★ If all the numbers are divided by a, then their average must be divided by a

Ex. 1 Find out the average of 308, 125, 45, 120 and 102.

Sol. Required average $= \dfrac{\text{Sum of given Observations}}{\text{Number of Observations}}$

$$= \frac{308+125+45+120+102}{5}$$

$$= \frac{700}{5} = 140$$

Ex. 2 If the weight of A is 60 kg, weight of B is 45 kg and weight of C is 54 kg, then what is the average weight of three persons?

Sol. Required average $= \dfrac{60+45+54}{3} = \dfrac{159}{3} = 53\ kg$

Ex. 3 The average expenditure of Chandan in four days is ₹90. If his expenditures for the first three days are ₹100, ₹125 and ₹85 respectively, then what is the expenditure of Chandan for the fourth day?

Sol. Let expenditure for fourth day = ₹x

Then, Average expenditure $= \dfrac{\text{Sum of the expenditures of four days}}{4}$

$$\therefore \quad 90 = \frac{100+125+85+x}{4}$$

$$\Longrightarrow \quad 310 + x = 360$$

$$\Longrightarrow \quad x = 360 - 310 = ₹50$$

Important Formulae Related to Average of Numbers

1. Average of first n natural numbers $= \frac{n+1}{2}$
2. Average of first n even numbers $= (n+1)$
3. Average of first n odd numbers $= n$
4. Average of consecutive numbers $= \frac{\text{First Number+Second Number}}{2}$
5. Average of 1 to n odd numbers $= \frac{\text{Last odd number+1}}{2}$
6. Average of 1 to n even numbers $= \frac{\text{Last even number+2}}{2}$
7. Average of the squares of first n natural numbers $= \frac{(n+1)(2n+1)}{6}$
8. Average of the cubes of first n natural numbers $= \frac{n(n+1)^2}{4}$
9. Average of n multiples of any number $= \frac{\text{Number} \times (n+1)}{2}$

Ex. 4 What will be the average of numbers from 1 to 51?

Sol. According to the formula,

The average of first n natural numbers $= \frac{n+1}{2}$

Here, $n = 51$

$$\therefore \quad \text{Required average} = \frac{51+1}{2} = \frac{52}{2} = 26$$

Ex. 5 Find out the average of $2, 4, 6, 8, 10, 12$ and 14.

Sol. As we know, average of first n even numbers $= (n+1)$

$$\therefore \quad \text{Required average} = (7+1) = 8$$

Ex. 6 Calculate the average of $1, 3, 5, 7, 9, 11, 13, 15$ and 17.

Sol. As we know,

Average of first n odd numbers $= n$

Here, $n = 9$

$$\therefore \quad \text{Required average} = 9$$

Ex. 7 What will be the average of $1, 2, 3, \ldots, 51, 52, 53$?

Sol. As we know, average of consecutive numbers $= \dfrac{\text{First Number + Second Number}}{2}$

Here, First Number $= 1$ and the Last number $= 53$

$$\therefore \quad \text{Required Average} = \frac{1+53}{2} = \frac{54}{2} = 27$$

Ex.8 Find out the average of $4, 7, 10, 13, \ldots, 28, 31$

Sol. Here, the difference between any two numbers written in continuous sequence is 3. Hence, this is a series of consecutive numbers.

As we know, average of consecutive numbers $= \dfrac{\text{First Number + Second Number}}{2}$

Here, First Number $= 4$ and the Last number $= 31$

$$\therefore \quad \text{Required Average} = \frac{4+31}{2} = \frac{35}{2} = 17.5$$

Ex.9 Find the average of all the odd numbers and average of all the even numbers from 1 to 45.

Sol. According to the formula,

$$\text{Average of 1 to n odd numbers} = \frac{\text{Last odd number+1}}{2}$$

Here, the last odd number $= 45$

$$\therefore \quad \text{Average of 1 to 45 odd numbers} = \frac{45+1}{2} = \frac{46}{2} = 23$$

According to the formula,

$$\text{Average of 1 to n even numbers} = \frac{\text{Last even number+2}}{2}$$

Here, the last even number $= 44$

$$\therefore \quad \text{Average of 1 to 45 even numbers} = \frac{44+2}{2} = \frac{46}{2} = 23$$

Ex. 10 Calculate the average of the squares of natural numbers from 1 to 25.

Sol. According to the formula,

$$\text{Average of the squares of first } n \text{ natural numbers} = \frac{(n+1)(2n+1)}{6}$$

Here, $n = 25$

$$\therefore \quad \text{Required Average} = \frac{(25+1)(2\times 25+1}{6} = \frac{26\times 51}{6} = 221$$

Ex. 11 Calculate the average of the cubes of first 9 natural numbers.

Sol. According to the formula,

$$\text{Average of the cubes of first } n \text{ natural numbers } = \frac{n(n+1)^2}{4}$$

Here, $n = 9$

$$\therefore \quad \text{Required Average } = \frac{9(9+1)^2}{4} = \frac{9 \times 100}{4} = 225$$

Ex. 12 What will be the average of first 9 multiples of 5?

Sol. According to the formula,

$$\text{Average of } n \text{ multiples of any number } = \frac{\text{Number } \times (n+1)}{2}$$

Here, $n = 9$ and Number = 5

$$\therefore \quad \text{Required Average} = \frac{5 \times (9+1)}{2} = \frac{5 \times 10}{2} = 25$$

Fast Track Techniques to solve the QUESTIONS

Technique 1

If the average of n_1 observations is a_1, the average of n_2 observations is a_2 and so on, then

$$\text{Average of all the observations } = \frac{n_1 a_1 + n_2 a_2 + \ldots}{n_1 + n_2 + \ldots}$$

Ex. 13 There are 30 boys and 60 girls in a class. If the average age of boys is 12 yr and average age of girls is 10 yr, then find out the average age of the whole class.

Sol. Here, $n_1 = 30, n_2 = 60, a_1 = 12, a_2 = 10$

$$\text{Average age of the whole class } = \frac{n_1 a_1 + n_2 a_2}{n_1 + n_2}$$

$$= \frac{30 \times 12 + 60 \times 10}{30 + 60} = \frac{360 + 600}{90} = 10.66 \text{ yr}$$

Ex.14 Pinky bought 20 books at the rate of ₹10 each, 45 pens at the rate of ₹5 each and 15 pencils at the rate of ₹3 each. Calculate the average price of all the stationary goods.

Sol. Here, $n_1 = 20, n_2 = 45, n_3 = 15, a_1 = 10, a_2 = 5, a_3 = 3$

$\therefore$ Average price of all the stationary goods $= \frac{n_1 a_1 + n_2 a_2 + n_3 a_3}{n_1 + n_2 + n_3}$

$$= \frac{20 \times 10 + 45 \times 5 + 15 \times 3}{20 + 45 + 15}$$
$$= \frac{200 + 225 + 45}{80} = ₹5.875$$

Technique 2

If the average of m observations is a and the average of n observations taken out of m is b, then

$$\text{Average of rest of the observations } = \frac{ma - nb}{m - n}$$

Ex.15 A man bought 20 cows in ₹200000. If the average cost of 12 cows is ₹12500, then what will be the average cost of remaining cows?

Sol. Total cost of 20 cows = 200000

Average cost of 12 cows = ₹12500

Total cost of 12cows = $12 \times 12500 = ₹150000$

Cost of remaining 8 cows = $200000 - 150000 = ₹50000$

$\therefore$ Average cost of remaining 8 cows $= \frac{50000}{8} = ₹6250$

Fast Track Method

$$\text{Average cost of 20 cows } = \frac{200000}{20} = ₹10000$$

Here, $m = 20,\ n = 12,\ a = 10000,\ b = 12500$

$$\therefore \text{ Average cost of remaining cows } = \frac{20 \times 10000 - 12 \times 12500}{20 - 12}$$
$$= \frac{200000 - 150000}{8} = \frac{50000}{8} = ₹6250$$

Technique 3

(i) if average of n observations is a but, the average becomes b when one observation is eliminated,then

$$\text{Value of eliminated observation } = n(a - b) + b$$

(ii) if average of n observations is a but the average becomes b when a new observation is added, then

$$\text{Value of added observation } = n(b - a) + b$$

Ex. 16 In a cricket team, the average age of 11 players and the coach is 18 yr. If the age of the coach is not considered, then the average decreases by 1 yr. Find out the age of the coach.

Sol. Total age of 11 players and the coach $= 12 \times 18 = 216$ yr

$$\text{Total age of 11 players} = 11 \times 17 = 187 \text{ yr}$$
$$\text{Age of the coach} = 216 - 187 = 29 \text{ yr}$$

Fast Track Method

$$\text{Here, } n = 11 + 1 = 12$$
$$\text{Initial average } (a) = 18 \text{ yr; Last average } (b) = 18 - 1 = 17 \text{ yr}$$
$$\therefore \quad \text{Age of the coach} = n(a-b) + b$$
$$= 12(18-17) + 17 = 12 + 17 = 29 \text{ yr}$$

Ex. 17 The average run scored by a batsman in 20 innings is 32. After 21st innings, the runs average becomes 34. How much runs does the batsman score in his 21st innings?

Sol.
$$\text{Runs scored in 20 innings} = 20 \times 32 = 640$$
$$\text{Runs scored in 21 innings} = 21 \times 34 = 714$$
$$\text{Runs scored in the 21st innings} = 714 - 640 = 74$$

Fast Track Method

$$\text{Here, } n = 20$$
$$\text{Initial average}(a) = 32; \text{Last average}(b) = 34$$
$$\text{Runs scored in 21st innings} = n(b-a) + b$$
$$= 20(34-32) + 34 = 20 \times 2 + 34 = 74$$

Technique 4

We have n observations out of which some observations $(a_1,\ a_2\ , a_3, \ldots)$ are replaced by some other new observations and in this way, if the average increases or decreases by b, then

$$\text{Value of new observations} = a \pm nb \qquad \text{where, } a = a_1 + a_2 + a_3 + \ldots$$

Note: In this formula, the signs of '+' and '-' depend upon the increment or decrement in the average.

Ex. 18 The average weight of 3 women is increased by 4 kg, when one of them whose weight is 100 kg, is replaced by another woman. What is the weight of the new woman?

Sol.
$$\text{Total weight increased} = 4 \times 3 = 12 \text{ kg}$$
$$\text{Weight of new woman} = 100 + 12 = 112 \text{ kg}$$

Fast Track Method

$$\text{Here, } n = 3, \quad a = 100 \text{ kg}, \quad b = 4 \text{ kg}$$
$$\text{Weight of new woman} = a + nb = 100 + 3 \times 4 = 112 \text{ kg}$$

[here, '+' sign has been taken as average increases in this case.]

Ex. 19 The average age of 25 boys in a class decreases by 6 months, when a new boy takes the place of a 20 yr old boy. Find out the age of new boy.

Sol. As the average age is decreased by 6 months $i.e.$, $\frac{1}{2}$yr.

Total age decreased $= 25 \times \frac{1}{2} = 12.5$yr

$\therefore$ Age of new boy = (Age of boy replaced -Total age decreased)

$= 20 - 12.5 = 7.5$yr

Fast Track Method

Here, $n = 25, a = 20, b = 6$ months $= \frac{6}{12} = 0.5$yr

$\therefore$ Age of the new boy $= a - nb$ [here, '-' sign has been taken as average increases in this case.]

$= 20 - 25 \times 0.5 = 20 - 12.5 = 7.5$ yr

Technique 5

If the average of n students in a class is a, where average of passed students is x and average of failed students is y, then

$$\text{Number of students passed} = \frac{\text{Total Students(Total Average - Average of failed students)}}{\text{Average of passed students - Average of failed students}}$$

$$= \frac{n(a-y)}{(x-y)}$$

Ex.20 In a class, there are 75 students and their average marks in the annual examination is 35. If the average marks of passed students is 55 and average marks of failed students is 30, then find out the number of students who failed.

Sol. Let number of failed students $= x$

Then, number of passed students $= 75 - x$

According to the question,

$$75 \times 35 = 30x + 55(75 - x)$$

$$\Longrightarrow \quad 15 \times 35 = 6x + 11(75 - x)$$

$$\Longrightarrow \quad 525 = 6x + 11x + 825$$

$$\Longrightarrow \quad 5x = 300$$

$$\therefore \quad x = 60$$

Fast Track Method

Here, $n = 75, a = 35, x = 55, y = 30$

$\therefore$ Number of students passed $= \dfrac{n(a-y)}{x-y}$

$= \dfrac{75(35-30)}{55-30} = \dfrac{75\times 5}{25} = 15$

$\therefore$ Number of students failed $= 75 - 15 = 60$

Technique 6

If the average of total components in a group is a, where average of n components (1st part) is b and average of remaining components (2nd part) is c, then

$$\text{Number of remaining components(2nd part)} = \frac{n(a-b)}{(c-a)}$$

Ex.21 The average salary of the entire staff in an office is ₹200 per day. The average salary of officers is ₹550 and that of non-officers is ₹120. If the number of officers is 16, then find the numbers of non-officers in the office.

Sol. Let number of non-officers $= x$

Then, $120x + 550\times 16 = 200(16+x)$

$\Longrightarrow$ $12x + 55\times 16 = 20(16+x)$

$\Longrightarrow$ $3x + 55\times 4 = 5(16+x)$

$\Longrightarrow$ $3x + 220 = 80 + 5x$

$\Longrightarrow$ $5x - 3x = 220 - 80$

$\Longrightarrow$ $2x = 140$

$\Longrightarrow$ $x = \dfrac{140}{2} = 70$

Fast Track Method

Here, $n = 16, a = 200, b = 550, c = 120$

According to the formula,

$$\text{Number of non officers} = \frac{n(a-b)}{c-a}$$

$$= \frac{16(200-550)}{120-200}$$

$$= \frac{16\times(-350)}{-80} = \frac{16\times 35}{8} = 70$$

Note Here, we have taken the office as a mixture of officers and non-officers

7.2 Average Speed

Average speed is defined as total distance travelled divided by total time taken.

$$\text{Average Speed} = \frac{\text{Total distance travelled}}{\text{Total time taken}}$$

Case I

If a person covers a certain distance at a speed of A km/h and again covers the same distance at a speed of B km/h, then

$$\text{The average speed during the whole journey} = \frac{2AB}{A+B}$$

Ex. 22 A person covers a certain distance by car at a speed of 25 km/h and comes back at a speed of 40 km/h. What is his average speed during his travel?

Sol. Here, $A = 25\ km/h$ and $B = 40\ km/h$

$$\therefore \quad \text{Required average speed} = \frac{2AB}{A+B} = \frac{2\times 25\times 40}{25+40} = \frac{2000}{65} = 30.76\ km/h$$

Case II

If a person covers three equal distance at the speed of $A\ km/hr$, $B\ km/hr$ and $C\ km/hr$ respectively, then

$$\text{Average speed during the whole journey} = \frac{3ABC}{AB+BC+CA}$$

Ex.23 If a person covers three equal distances at the speed of 30 km/h, 15 km/h and 10 km/h respectively, then find out his average speed during the whole journey.

Sol. Here, $A = 30\ km/h, B = 15\ km/h$ and $C = 10\ km/h$

$$\therefore \quad \text{Required average speed} = \frac{3ABC}{AB+BC+CA} = \frac{3\times 30\times 15\times 10}{30\times 15+15\times 10+10\times 30}$$
$$= \frac{3\times 30\times 15\times 10}{450+150+300} = \frac{3\times 30\times 15\times 10}{900} = 15\ km/hr$$

Case III

If distance P is covered with speed x, distance Q is covered with speed y and distance R is covered with speed z and so on. Then, for the whole journey,

$$\text{Average Speed} = \frac{P+Q+R+\ldots}{\frac{P}{x}+\frac{Q}{y}+\frac{R}{z}+\ldots}$$

Ex. 24 A person covers 20 km distance with a speed of 5 km/h, then he covers the next 15 km with a speed of 3 km/h and the last 10 km is covered by him with a speed of 2 km/h. Find out his average speed for the whole journey.

Sol. Here, $P = 20\ km,\ Q = 15\ km, R = 10\ km$

and $x = 5km/hr,\ y = 3\ km/hr,\ z = 2\ km/hr$

$$\therefore \quad \text{Required average speed } = \frac{P+Q+R+\ldots}{\frac{P}{x}+\frac{Q}{y}+\frac{R}{z}+\ldots} = \frac{20+15+10}{\frac{20}{5}+\frac{15}{3}+\frac{10}{2}}$$

$$= \frac{45}{4+5+5} = \frac{45}{14} = 3\frac{3}{14}\ km/hr$$

Case IV

If a person covers P part of his total distance with speed of x, Q part of total distance with speed of y and R part of total distance with speed of z, then

$$\text{Average Speed } = \frac{1}{\frac{P}{x}+\frac{Q}{y}+\frac{R}{z}}$$

Ex.25 Mr. Sharma travels by car and covers 25% of his journey with a speed of 10 km/h, 45% of his journey with a speed of 5 km/h and remaining 30% of his journey with a speed of 15 km/ h. What will be the average speed of Mr. Sharma for the whole journey?

Sol. Here, $P = 25\% = \frac{1}{4}\ ;\ x = 10\ km/hr$

$$Q = 45\% = \frac{9}{20}\ ;\ y = 5\ km/hr$$

$$R = 30\% = \frac{3}{10}\ ;\ z = 15\ km/hr$$

$$\therefore \quad \text{Required Average Speed } = \frac{1}{\frac{P}{x}+\frac{Q}{y}+\frac{R}{z}}$$

$$= \frac{1}{\frac{1}{4\times10}+\frac{9}{20\times5}+\frac{3}{10\times15}} = \frac{1}{\frac{1}{40}+\frac{9}{100}+\frac{1}{50}}$$

$$= \frac{1}{\frac{5+18+4}{200}} = \frac{200}{27} = 7.40\ km/hr$$